14 **A** $\frac{P_1V_1}{n_1T_1} = \frac{P_2V_2}{n_2T_2} \Rightarrow \frac{P_1V_1}{m_1T_1} = \frac{P_2V_2}{m_2T_2}$ since $n = \frac{mass}{molar\ \text{mass}}$

As pressure and volume remains constant,

$m_1T_1 = m_2T_2$

$(2.0)(300) = m_2(400)$

$m_2 = 1.5$

Hence mass of gas escaping = 2.0 - 1.5 = 0.50 kg

15 **C** As temperature is constant, the internal energy of the gas is unchanged